

Lecture 9: Memory Systems (MCQ Questions)

1. The memory hierarchy system consists of several levels, where the top level (Registers) is characterized by being _____. a) Slowest and cheapest b) Fastest and most expensive c) Large capacity and volatile d) Non-volatile and cheap

• **Answer:** b) Fastest and most expensive

2. Which of the following is the time interval between the request for information and the access to the first bit of that information? a) Cycle Time b) Latency (Access Time) c) Bandwidth d) Transfer Rate

• **Answer:** b) Latency (Access Time), (مكرر في امتحان 2025)

3. The time elapsed from the start of a read operation to the start of a subsequent read is called _____. a) Access Time b) Seek Time c) Cycle Time d) Bandwidth

• **Answer:** c) Cycle Time, (مكرر في امتحان 2025)

4. In Random Access Memory (RAM), the Cycle Time is usually _____ the Access Time. a) Less than b) Equal to c) Greater than d) Unrelated to

• **Answer:** c) Greater than (Because Cycle time = Access time + Recovery time)

5. Which type of memory requires periodic refreshing to maintain its data? a) SRAM b) DRAM c) ROM d) Flash Memory

• **Answer:** b) DRAM

6. Static RAM (SRAM) is constructed using _____. a) Capacitors b) Flip-flops c) Magnetic tapes d) Transistors only

• **Answer:** b) Flip-flops

7. Which of the following statements is TRUE about SRAM compared to DRAM? a) SRAM is cheaper b) SRAM is slower c) SRAM does not need refresh d) SRAM has higher density

• **Answer:** c) SRAM does not need refresh

8. Because of its high speed and cost, SRAM is typically used for _____. a) Main Memory b) Hard Disk c) Cache Memory d) Off-line storage

• **Answer:** c) Cache Memory

9. DRAM is typically used for constructing _____ because it is cheaper and denser. a) Registers b) L1 Cache c) Main Memory d) Firmware

• **Answer:** c) Main Memory

10. A memory that loses its content when the power is turned off is called _____. a) Non-volatile b) Volatile c) Magnetic d) Read-Only

• **Answer:** b) Volatile,

11. Which type of ROM can be erased and reprogrammed by exposing it to Ultraviolet (UV) light? a) PROM b) EPROM c) EEPROM d) Mask ROM

• **Answer:** b) EPROM

12. EEPROM stands for _____. a) Electrically Erasable Programmable Read Only Memory b) Erasable External Programmable Read Only Memory c) Energy Efficient Programmable Read Only Memory d) Easy Erasable Programmable Read Only Memory

• **Answer:** a) Electrically Erasable Programmable Read Only Memory

13. Which memory type can be programmed only once? a) EPROM b) EEPROM c) PROM d) Flash Memory

• **Answer:** c) PROM

14. In memory organization, if the number of address lines is , the number of memory locations (rows) is _____. a) b) c) d)

• **Answer:** c)

15. A given in the memory matrix is used to enable _____. a) All cells in a column b) All cells in a row c) One single cell d) The address decoder

• **Answer:** b) All cells in a row

16. A given is common to all cells in a given _____. a) Row b) Block c) Column d) Chip

• **Answer:** c) Column

17. If we want to design a 4M Byte memory using 1M Bit chips, how many chips do we need? (Calculation: Total Size = 4M 8 bits = 32M bits. Chip Size = 1M bits. Count = $32/1$) a) 4 Chips b) 8 Chips c) 16 Chips d) 32 Chips

• **Answer:** d) 32 Chips

18. The "Start-up" or "Boot" program of a computer is usually stored in _____. a) RAM b) ROM c) Cache d) CD-ROM

• **Answer:** b) ROM

19. Which of the following has the cost per bit? a) Registers b) Cache c) Main Memory d) Magnetic Tape

• **Answer:** d) Magnetic Tape,

20. Access Time is constant for all locations in _____. a) Magnetic Tape b) Magnetic Disk c) Random Access Memory (RAM) d) Optical Disk

• **Answer:** c) Random Access Memory (RAM)

21. _____ is a measure of the number of bits per second that can be accessed. a) Latency b) Capacity c) Bandwidth d) Frequency

• **Answer:** c) Bandwidth

22. Which component activates only one word select line based on the input address? a) Multiplexer b) ALU c) Address Decoder d) Data Register

• **Answer:** c) Address Decoder

23. Flash memory is a specific type of _____. a) SRAM b) DRAM c) EEPROM d) Mask ROM

• **Answer:** c) EEPROM

24. The L1 cache is _____ than the L3 cache. a) Slower b) Faster c) Same speed d) Larger

• **Answer:** b) Faster

25. A memory organized as has _____ address lines. (Calculation:) a) 4 b) 8 c) 10 d) 12

• **Answer:** c) 10

26. Which of the following is considered "Main Memory"? a) Hard Disk b) DRAM c) CD-ROM d) Flash Drive

• **Answer:** b) DRAM

27. To erase an EPROM, the chip must be _____. a) Removed from the circuit b) Connected to high voltage c) Refreshed d) Formatted by software

• **Answer:** a) Removed from the circuit

28. Which characteristic increases as we move the memory hierarchy (from Registers to Tape)? a) Cost per bit b) Access Speed c) Capacity (Size) d) Frequency of CPU access

• **Answer:** c) Capacity (Size),

29. The contents of _____ are lost if the power is turned off. a) EPROM b) ROM c) RAM d) Flash Memory

• **Answer:** c) RAM

30. In the memory hierarchy, the memory level directly accessible by the CPU (after registers and cache) is _____. a) Magnetic Disk b) Main Memory c) Magnetic Tape d) Optical Disk

• Answer: b) Main Memory